

Gowthaman Aravindan

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Experience

Shopee (Payments)

Jul 2025 – Present

Software Engineer

Singapore

- Architected an **ML-driven** payment method ranking system using dual LightGBM models to predict user preference and payment success, with a versioned MySQL feature store and A/B experimentation, matching users' final payment method choices in over **80%** of checkouts.
- Designed ShopeePay's **offline QR payment protocol** with HMAC-authenticated payment tokens, Redis-backed distributed locks, idempotent transactions, and a Saga-based state machine, enabling secure internet-free payments for **1–2M** users across **six regions**.
- Built a **multi-agent AI** system to triage payment reconciliation alerts and automate root-cause analysis, reducing on-call effort by **~20%**.
- Developed APIs for third-party QR payment integrations (Mitra Shopee), supporting **~9.3K** merchants and generating **~16K** daily orders.
- Collaborated with regional product teams and payment networks to launch Buy Now, Pay Later (BNPL) capabilities across Indonesia and the Philippines, including automatic credit refresh, mobile-device insurance payments, referral tracking, and QR-based account activation.
- Re-architected order sync from binlog-based **CDC** to real-time APIs with **Kafka**-backed retries, preventing stale merchant dashboard data.
- Reduced financial data resync time by **80%** through bounded concurrent queries across database shards, sustaining over **4K** peak QPS.

GovTech (Search)

Aug 2023 – Jan 2024

Software Engineer Intern

Singapore

- Revamped DOM parsing and visibility heuristics in the **crawler** and **indexer**, recovering **3,000+** lost metadata fields across **20+** domains.
- Integrated **AWS QuickSight** dashboards with secure embedding, reducing load times from **5s to <1s** by serving precomputed analytics.
- Strengthened **Express.js** auth middleware by replacing query-based validation with **JWT** claims, enforcing multi-tenant access control.
- Developed stateless login workflows using **JWT cookies**, enabling persistent user sessions without needing server-side session storage.

Google (Payments)

May 2023 – Aug 2023

Software Engineer Intern

Singapore

- Built a **Java**-based UI tool to surface accessibility diffs via annotated screenshots in merge requests, reducing code review time by **20%**.
- Implemented a **gRPC** pipeline with custom **Protocol Buffer** schemas to automate the extraction of test results from a central database.
- Provisioned and maintained a **gRPC server**, ensuring continuous system reliability through performance monitoring and alert handling.

Zendesk (Chat)

Jan 2023 – May 2023

Software Engineer Intern

Singapore

- Migrated developer environments from **AWS EC2** to **Kubernetes**, reducing boot time by **70%** and cutting monthly infra costs by **~\$5,000**.
- Engineered **GraphQL** mutations in **Java** to route client requests to upstream microservices, ensuring reliability through **JUnit** coverage.

Kinexcs (HealthTech)

Nov 2022 – Jan 2023

Software Engineer Intern

Singapore

- Implemented gamified rehabilitation exercises using **Flutter UI** components and **Unity**, resulting in a **40%** increase in patient engagement.
- Modeled and deployed scalable **DynamoDB** and **RDS** databases to store user profiles and manage application state for **10,000+** users.
- Developed backend services in **Python (Flask)** and containerized test environments using **Docker** to automate end-to-end API testing.

Projects

Pathfinding Visualizer [TypeScript]

Real-time, terrain-aware visualizer enabling side-by-side execution of **A***, **Dijkstra's**, and **BFS** on weighted grids via a custom traversal engine.

Education

National University of Singapore

Aug 2020 – Jan 2025

Bachelor of Engineering (Second Major in Computer Science)

Honors (Distinction)

Technical Skills

Languages: Go, Python, C++, TypeScript, JavaScript, Java

Technologies/Frameworks: React.js, Node.js, Express.js, Flask, MySQL, MongoDB, Flutter, GraphQL, Jenkins

Developer Tools: Docker, Kubernetes, AWS, Vercel, Prisma, Serverless, Git, Linux, Postman